European Polymer Journal 233 (2025) 113952

Contents lists available at ScienceDirect

European Polymer Journal

journal homepage: www.elsevier.com/locate/europolj

Generation-dependent aggregation-induced emission from unconventional chromophore aggregation in phosphonate-functionalized Si-containing dendritic polymers

Zhihui Yang [a,b,1], Fanghao Sun [a,b,1], Weizhi Tian [c], Haobo Shi [c], Haifeng Lu [a,b,*]

[a] School of Chemistry and Chemical Engineering, Shandong University, Jinan 250100, PR China
[b] Key Laboratory of Special Functional Aggregated Materials & (Shandong University), Ministry of Education, PR China
[c] Beijing Xinghang Electromechanical Equipment Co., Ltd., Beijing 100074, PR China

ARTICLE INFO

Keywords:
Aggregation-induced emission
Fluorescence
Phosphonates
Dendrimer

ABSTRACT

Unconventional luminescent macromolecules have attracted significant interest for their superior biocompatibility, processability, and biodegradability. However, achieving multicolor emission through structural modulation remains a substantial challenge. In this study, a series of novel phosphonate-functionalized, silicon-containing dendritic macromolecules were synthesized by incorporating phosphonate groups into the molecular structure without modifying the silicon-based dendritic framework. The fluorescence intensity of these macromolecules increased markedly with higher polymer generations, revealing a unique generation-dependent aggregation-induced emission (AIE) phenomenon. The AIE properties were further validated by examining the effects of concentration and poor solvents on fluorescence behavior. Experimental results demonstrate that phosphonate functionalization enhances the rigidity of the molecular backbone and stabilizes fluorescent clusters, amplifying non-conjugated linkages and through-space conjugation (TSC) effects. Additionally, cytotoxicity assays confirmed the excellent biocompatibility of these materials. This study establishes phosphonate end-grafting as an effective strategy to optimize structural flexibility and achieve generation-dependent emission in non-conjugated polymers, paving the way for applications in bioimaging and fluorescent labeling while offering new insights into the design of advanced luminescent materials.

1. Introduction

While conventional fluorophores suffer from aggregation-caused quenching (ACQ) due to the formation of non-emissive clusters at high concentrations [1–3], the discovery of aggregation-induced emission (AIE) materials provided a transformative alternative by utilizing aggregation to enhance fluorescence [4–8]. AIE-active macromolecules, as a versatile class of soft luminescent materials, have garnered significant attention owing to their facile synthesis, excellent processability, high brightness in aggregated and solid states, multifunctionality, and environmental responsiveness, thus driving rapid advancements in this field [9–20].

However, the prevailing dependence on π-conjugated aromatic architectures in AIE systems introduces intrinsic constraints, particularly regarding environmental compatibility and emission wavelength programmability [21–26]. These limitations have catalyzed exploration of unconventional chromophoric systems incorporating non-aromatic motifs (e.g., aliphatic amines, P=O, C=N), which demonstrate superior biocompatibility but are hampered by mediocre quantum efficiencies (<10 %) due to inadequate excited-state stabilization [27–35]. Parallel developments in silicon-based photonic materials have revealed unique opportunities through σ*-π* hyperconjugation [36–44] and intramolecular charge transfer (ICT) effects enabled by silicon's distinctive electronic configuration [45–53]. Our foundational studies on Si-PAMAM dendrimers [54] elucidated the synergistic interplay between carbonyl aggregation and N → Si coordination, yet the precise mechanistic control required for emission wavelength engineering remained unresolved.

To address these limitations, we introduce a novel class of phosphonate-functionalized siloxane dendrimers (generations G0-G2)

* Corresponding author at: School of Chemistry and Chemical Engineering, Shandong University, Jinan 250100, PR China.
E-mail address: lhf@sdu.edu.cn (H. Lu).
[1] The two authors have equal contributions to this work.

https://doi.org/10.1016/j.eurpolymj.2025.113952
Received 9 February 2025; Received in revised form 10 April 2025; Accepted 15 April 2025
Available online 27 April 2025
0014-3057/© 2025 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

that exploit the cooperative interaction between electron-donating silicon centers and electron-withdrawing P=O groups. This Si-P synergy establishes a push–pull system that concurrently (i) stabilizes excited states through p-π conjugation, (ii) enhances ICT via orbital hybridization, and (iii) restricts molecular motion through steric confinement, resulting in generation-dependent AIE enhancement, the photoluminescence quantum yield (Φ_F) attains 22.14 %, and tunable bathochromic shifts ($\Delta\lambda = 26$ nm).

By systematically varying dendrimer architecture while maintaining P=O groups as the structural core, we demonstrate that controlled progression through dendrimer generations enhances steric confinement around the P=O centers, leading to generation-dependent modulation of AIE behavior. Our findings elucidate key structure–property relationships and reveal the critical role of Si-P cooperativity in these photophysical processes.

2. Results and discussion

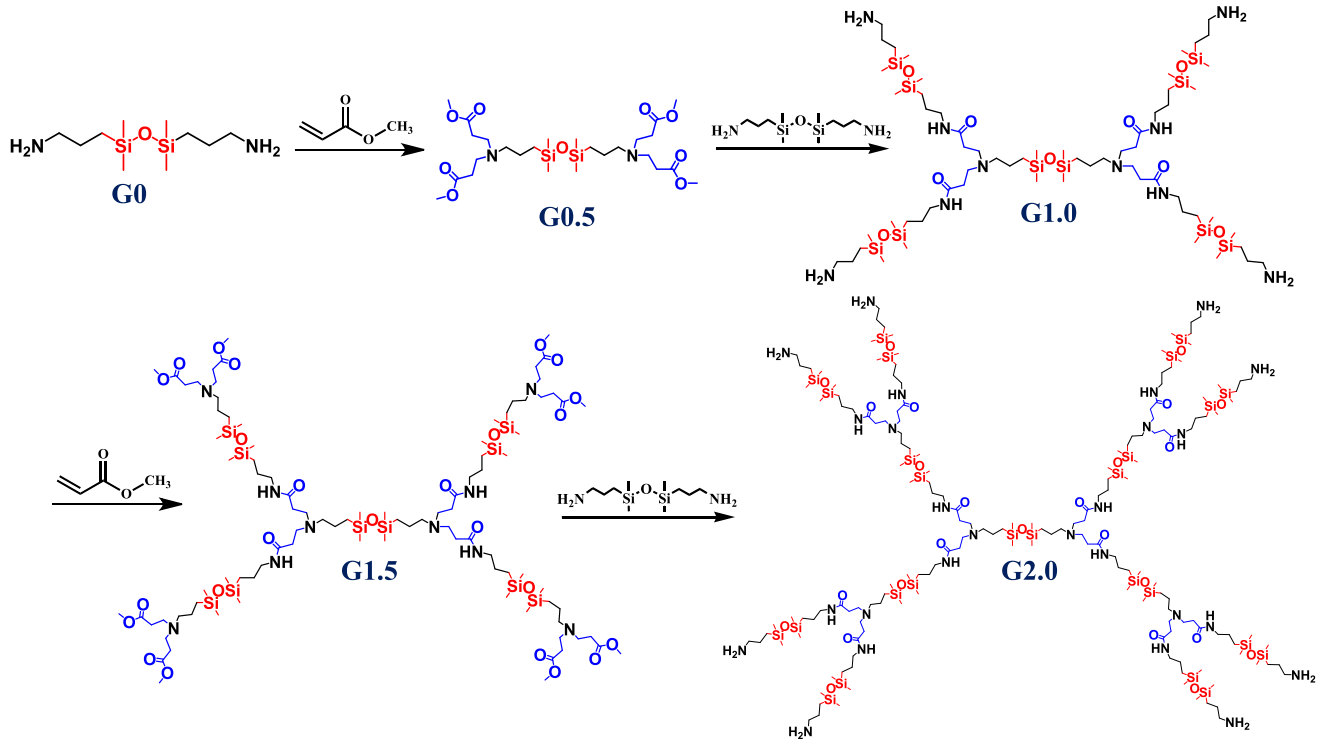
In this study, aiming to design and synthesize novel dendrimers exhibiting AIE properties, we innovatively introduced the phosphonate moiety as the core structure of the dendrimers. The synthesis process began with a commercially available chemical, 1,3-Bis(3-aminopropyl) tetramethyldisiloxane (labeled as G0), and through a series of chemical reactions, we successfully synthesized the first-generation (G1) and second-generation (G2) dendritic macromolecules. The synthetic pathway is shown in Scheme 1, and specific details of the synthesis are given in Supporting Materials.

Building on this foundation, phosphonate groups were strategically introduced into the frameworks of G0, G1, and G2, resulting in phosphonate-functionalized dendrimers G0-P, G1-P, and G2-P. Scheme 2 outlines the precise and controllable dispersion strategy used to synthesize these phosphonate-functionalized siloxane dendrimers. The general procedure involved dissolving G0, G1, and G2 in acetone and stirring the reaction mixture at room temperature. Diethyl phosphite and p-toluenesulfonic acid were then added to facilitate further reactions. Upon completion, most of the solvent was removed by rotary evaporation, followed by sequential washing with dichloromethane, saturated $NaHCO_3$, and NaCl solutions. The final steps included drying with Na_2SO_4, filtration, and rotary evaporation to obtain the yellow viscous liquid products G0-P, G1-P, and G2-P. Detailed synthesis procedures are provided in the Supporting Materials.

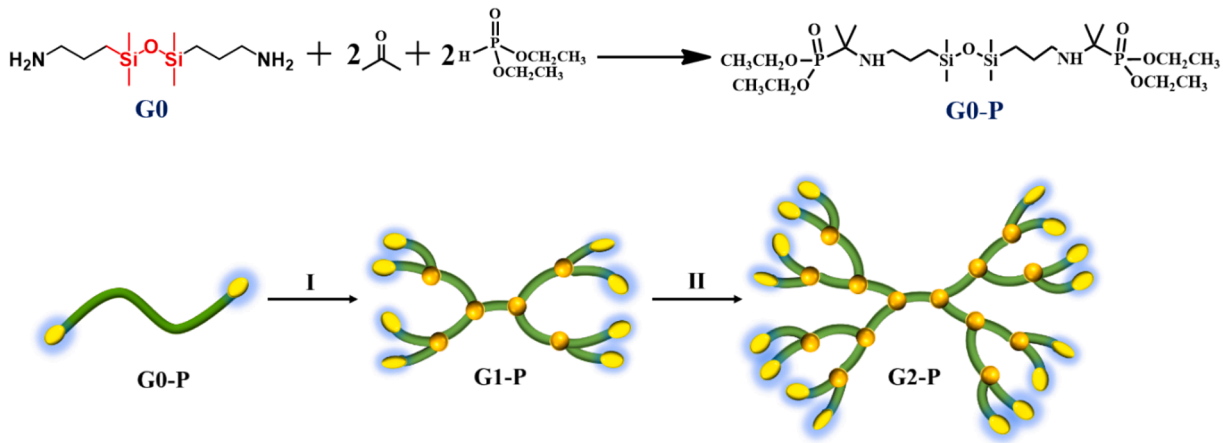
To comprehensively characterize the structural properties of the synthesized phosphonate-functionalized siloxane dendritic macromolecules, Fourier transform infrared spectroscopy (FT-IR), along with ^{1}H, ^{13}C, and ^{31}P nuclear magnetic resonance (NMR) spectroscopy, was performed, as shown in Figs. S1-S20. The data clearly confirm the structural features associated with phosphonate functionalization. Characteristic peaks corresponding to siloxane dendritic macromolecules were observed, affirming the incorporation of AIE-active luminescent groups. Notably, the ^{31}P NMR spectrum exhibited a single sharp peak attributed to the phosphonate group, which broadened progressively with increasing dendrimer generation, indicating subtle variations in the chemical environment of the phosphonate groups. Collectively, these results confirm the successful synthesis of phosphonate-functionalized siloxane dendrimers.

The comparative UV–vis absorption studies of phosphonate-functionalized (G0-P-G2-P, Fig. 1a) and non-functionalized (G0-G2) dendritic polymers in dilute CH_2Cl_2 solutions (Fig. S21) revealed significant electronic modifications induced by phosphonate incorporation. The parent dendrimers exhibited characteristic yet weak absorption bands at 235 nm (attributed to the carbonyl $\pi \rightarrow \pi^*$ transition) and 280 nm ($n \rightarrow \sigma^*$ transition) with minimal absorption (<0.2 AU) beyond 300 nm. In contrast, their phosphonate-functionalized counterparts showed three distinct modifications: (1) 5–8-fold enhancement in absorption intensity, (2) 15–30 nm bathochromic shifts in the $\pi \rightarrow \pi^*$ transition, and (3) emergence of new absorption features between 300–350 nm.

The observed spectral modifications are rationalized by three well-defined phosphonate-derived photophysical mechanisms: (i) delocalization enhancement through P=O π^*-orbital participation in the conjugated system, (ii) formation of distinct $n \rightarrow \pi^*$ transitions originating from phosphoryl non-bonding orbitals, and (iii) promoted charge



Scheme 1. Schematic representation of a controlled dispersion method for the synthesis of phosphonate-functionalized siloxane dendrimers G1 and G2.



Scheme 2. Schematic representation of a controlled dispersion method for the synthesis of phosphonate-functionalized siloxane dendrimers G1-P and G2-P.

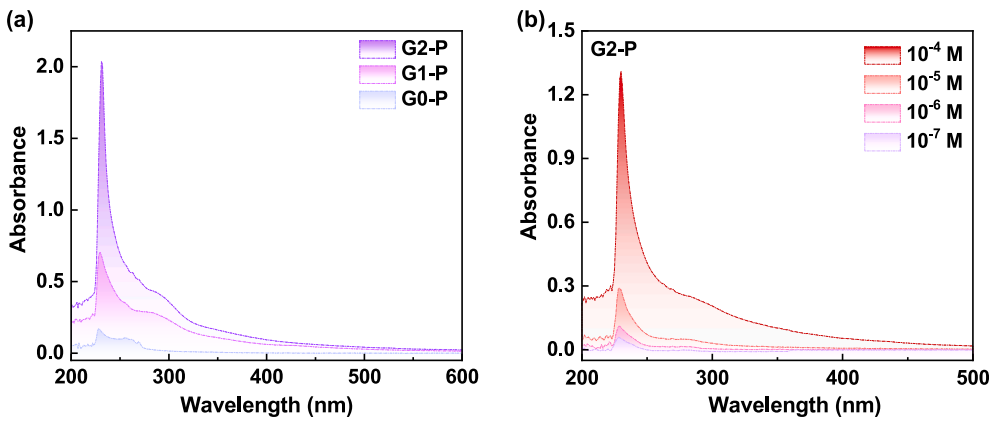


Fig. 1. (a) UV–vis absorption spectra of three compounds in CH_2Cl_2 solution ($c = 1 \times 10^{-5}$ M). (b) UV–vis absorption spectra of phosphonate-functionalized siloxane dendrimers (G2-P) at different concentrations in CH_2Cl_2.

separation via through-space dipole–dipole coupling of proximal phosphonate units. These generation-dependent photophysical changes, quantitatively characterized by progressive bathochromic shifts ($\Delta\lambda_{max}$ = 30 nm from G0-P to G2-P) with concomitant oscillator strength enhancement, provide some evidence for the bifunctional nature of phosphonate moieties: they simultaneously enable through-space electronic communication (TSC) within the dendritic scaffold while imposing structural constraints through intramolecular P=O···H-N hydrogen bond networks.

Concentration-dependent studies of G2-P in CH_2Cl_2 (Fig. 1b) further supported these conclusions, showing concentration-dependent absorption enhancement accompanied by a 10 nm bathochromic shift (230 nm → 240 nm). These results align with the generation-dependent trends, where increased branching and phosphonate functionalization

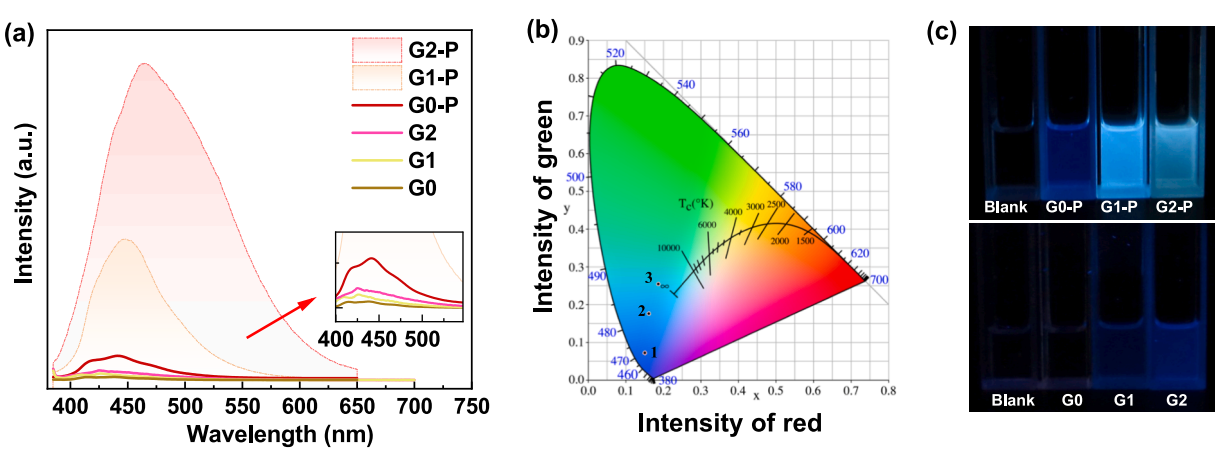


Fig. 2. (a) Emission spectra of G0, G1, G2, G0-P, G1-P and G2-P in CH_2Cl_2 at the same case (Excited at 375 nm, room temperature). (b) Emission colors of compounds in the CIE1931 chromaticity diagram (1: G0-P, 2: G1-P, 3: G2-P). (c) Images under UV illumination (λ_{ex} = 365 nm) of G0, G1, G2, G0-P, G1-P and G2-P. ($c = 1 \times 10^{-5}$ M).

work synergistically to enhance molecular rigidity and strengthen intramolecular interactions between amide and amino groups, ultimately resulting in the observed emission red-shifts.

The phosphonate moiety, serving as a core structural element, motivated us to further investigate the AIE properties of the synthesized dendritic macromolecules G0-P to G2-P. Solubility tests demonstrated that all siloxane-containing phosphonates dissolved readily in methanol. Consequently, methanol solutions of each compound at a concentration of 1×10^{-3} M were prepared, and their fluorescence intensities were measured. As shown in Fig. 2a, the phosphonate-functionalized siloxane dendrimers exhibited a substantial increase in fluorescence intensity and a noticeable redshift in emission wavelengths compared to unmodified siloxane dendrimers. Specifically, the maximum emission wavelength (λ_{emmax}) of G0-P, G1-P, and G2-P were 431 nm, 448 nm, and 457 nm, respectively. The observed bathochromic shift in emission maxima from 431 nm (G0-P) to 457 nm (G2-P) demonstrates a generation-dependent enhancement of excited-state dipole moment ($\Delta\mu$), which is diagnostic of ICT stabilization. This phenomenon arises from the synergistic interplay between electron-deficient phosphonate moieties (P=O) as acceptor units and electron-rich siloxane dendritic architectures as donor components.

Using the fluorescence data, we calculated the theoretical chromaticity coordinates of G0-P, G1-P, and G2-P on the CIE 1931 chromaticity diagram. As depicted in Fig. 2b, the CIE tristimulus values for G0-P, G1-P, and G2-P were (0.15, 0.07), (0.17, 0.18), and (0.21, 0.25), respectively, accurately representing the emission color positions. The progression of luminescence colors is clearly illustrated in the chromaticity diagram. Fig. 2c presents photographs of the six compounds, along with a blank reference sample, under 365 nm UV light. The phosphonate-modified siloxane dendrimers exhibited varying brightness levels visible to the naked eye. These results confirm that phosphonate functionalization significantly enhances both the luminescence intensity and color modulation of the siloxane dendrimers, demonstrating a pronounced dimming effect on luminescence color.

To further investigate their AIE behavior, we varied the concentrations of G0-P, G1-P, and G2-P in methanol and monitored the impact on luminescence intensity. The results demonstrated a clear trend of increasing fluorescence intensity with higher concentrations for all three materials. Additionally, this increase in fluorescence was accompanied by a redshift in emission wavelengths, as shown in Fig. 3a-3c. These findings confirm a strong correlation between solute concentration and AIE properties, highlighting how concentration influences both emission intensity and wavelength modulation in phosphonate-functionalized siloxane dendritic macromolecules.

Capitalizing on these findings, we conducted a systematic investigation of generation-dependent photophysical properties in methanol/water systems, exploiting the controlled solubility gradients of our phosphate-functionalized dendritic polymers. Through rigorous use of ultrapure water to preclude ionic artifacts, Fig. 3d-3f reveal a striking dendrimer-generation-correlated fluorescence amplification upon water fraction increase. Although this solvatochromic enhancement bears superficial resemblance to AIE phenomenology, the distinctive supramolecular organization of these dendritic chromophores-arising from their architecturally constrained aggregation pathways – implies a non-canonical photophysical mechanism requiring deeper exploration. Collectively, the G0-P to G2-P systems constitute a novel category of structurally engineered luminescent materials whose emission is exquisitely modulated by hierarchical aggregation states.

Given the variation in steric hindrance across different generations of phosphonate-functionalized siloxane dendrimers, we hypothesized that increased viscosity would enhance the restriction of intramolecular rotation (RIR), thereby significantly affecting their AIE performance. To investigate this generation-dependent behavior, we studied compounds G0-P through G2-P. Glycerol, with its high viscosity, was selected to restrict free molecular motion and explore molecular dynamics.

Systematic fluorescence studies in methanol-glycerol mixtures

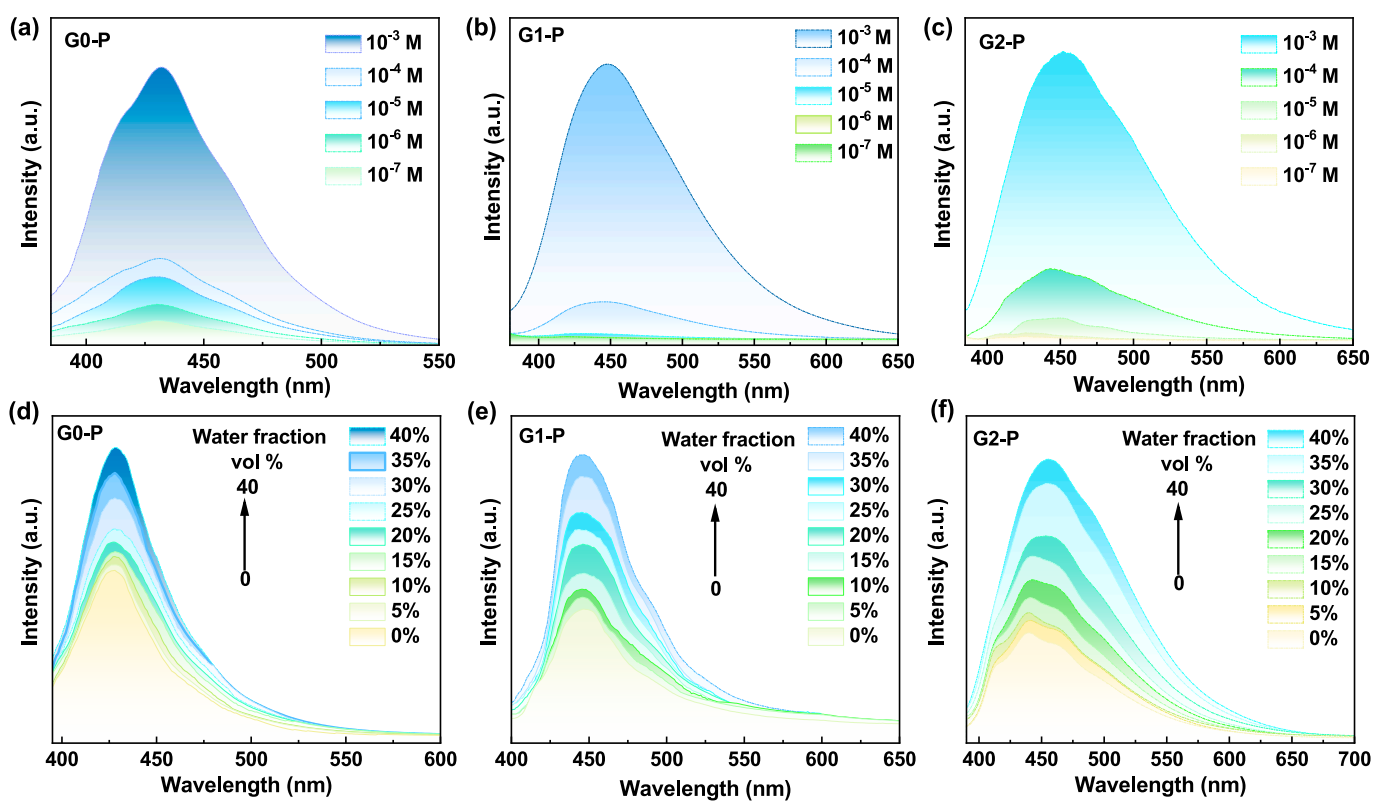


Fig. 3. (a-c) Emission spectra of G0-P, G1-P and G2-P with different concentrations in CH_2Cl_2, λ_{ex} = 375 nm. (d-f) Emission spectra of MeOH/H_2O solutions containing with increasing H_2O fractions. λ_{ex} = 375 nm, $c = 1 \times 10^{-5}$ M.

revealed distinct generation-dependent photophysical behavior. Fig. 4a-4c demonstrate a consistent enhancement in fluorescence intensity with increasing glycerol content. While G0-P maintained its emission maximum without significant redshift even at 99 % glycerol, G1-P exhibited a notable 11 nm blue shift (447 → 436 nm) and G2-P showed a more pronounced 17 nm shift (452 → 435 nm) under the same conditions. This spectral evolution can be attributed to the dramatic viscosity increase in glycerol-rich media, which effectively suppresses intramolecular motions (particularly rotational and vibrational modes) of the AIE-active chromophores. The resulting restriction of molecular dynamics preferentially inhibits non-radiative decay pathways while promoting radiative transitions, accounting for both the observed emission enhancement and hypsochromic shifts. Notably, the generation-dependent magnitude of these effects correlates with the degree of molecular rigidity imposed by the dendritic architecture. These findings provide evidence that the AIE behavior in phosphonate-functionalized siloxane dendrimers originates primarily from restricted intramolecular motion, with higher-generation dendrimers exhibiting more pronounced effects due to their greater structural constraints.

The phosphonate-functionalized dendrimers (G0-P to G2-P) demonstrate distinct donor–acceptor (D-A) properties, with the silicon framework donating electrons to P=O acceptor groups, enabling efficient ICT. Solvent polarity studies (Fig. 4d-f) reveal that while high-polarity media stabilize ICT states, they simultaneously promote non-radiative decay through enhanced molecular vibrations. This effect shows generation-dependence: G0-P exhibits minimal polarity sensitivity due to limited D-A separation, whereas G1-P/G2-P display significantly enhanced responses through improved conjugation and through-space charge transfer. The characteristic 17 nm chromic shift in viscous glycerol (G2-P, 452 → 435 nm) confirms restricted ICT formation due to hindered molecular rotation. These results establish that luminescence in these systems balances polarity-driven ICT stabilization against viscosity-mediated molecular rigidity, providing fundamental insights for designing environment-responsive AIE materials.

Quantum yield (Φ_F) measurements provided further corroboration of these observations. As summarized in Table 1, the fluorescence lifetimes exhibited a distinct generational progression, increasing from 1.90 ns (G0-P) to 3.45 ns (G1-P) and 6.40 ns (G2-P). The biexponential decay profiles suggest the coexistence of multiple radiative decay pathways. Notably, the systematic enhancement of fluorescence lifetime across generations aligns well with the proposed TSC-mediated exciton delocalization mechanism. Furthermore, the quantum yields demonstrated a pronounced generation-dependent enhancement, escalating from 4.67 % for G0-P to 12.32 % for G1-P and reaching 22.14 % for G2-P (λ_{ex} = 375 nm). This progressive improvement in quantum efficiency strongly correlates with the increasing density of the three-dimensional TSC network formed at higher dendrimer generations. These findings collectively demonstrate that increased phosphonate aggregation amplifies the AIE effect, in agreement with Yang et al.'s report [55] which established that larger nano-aggregates in higher-generation dendrimers produce more pronounced AIE characteristics.

To further validate the AIE phenomenon, dynamic light scattering (DLS) and transmission electron microscopy (TEM) analyses were performed on G0-P to G2-P. Fig. 5a presents the DLS characterization results, revealing a distinct generation-dependent progression in hydrodynamic size upon addition of 60 % water (a poor solvent), with Z-average diameters increasing sequentially from 3.9 nm (G0-P) to 39.4 nm (G1-P) and finally to 89.5 nm (G2-P). This systematic molecular expansion, combined with the spectroscopic evidence presented in Figs. 3-4, strongly suggests that higher-generation architectures enhance chromophore packing density through through-space conjugation (TSC)-mediated spatial confinement effects.

TEM images (Figs. S22-S24) demonstrate generation-dependent aggregation in 60 % aqueous solution: (i) G0-P remains dispersed, (ii) G1-P forms monodisperse nanoparticles (38.7 ± 2.1 nm), and (iii) G2-P assembles into three-dimensional dendritic networks (89.5 ± 8.1 nm).

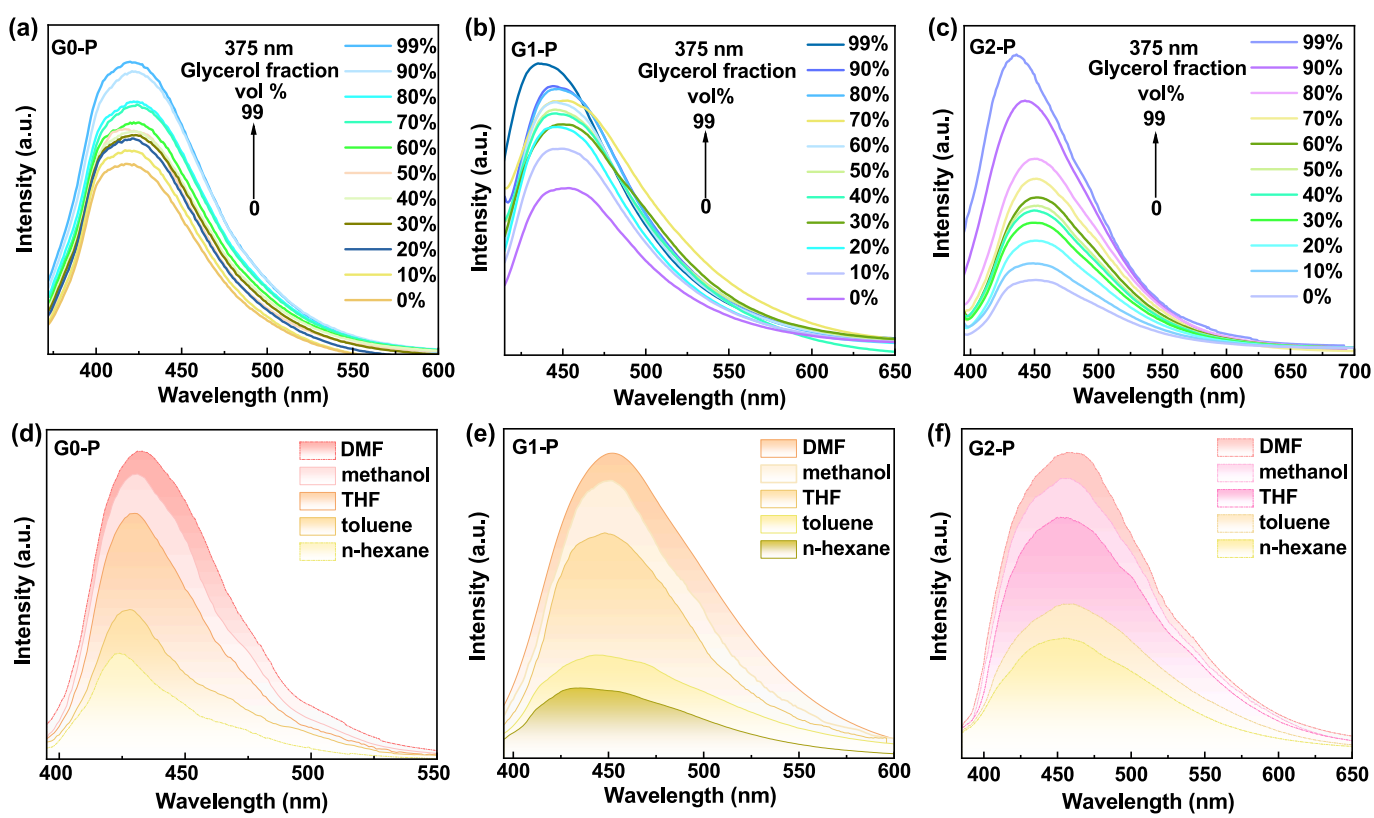


Fig. 4. (a-c) Fluorescence spectra of G0-P, G1-P and G2-P with different concentrations in methanol/ glycerin solution, (d-f) Emission spectra of G0-P, G1-P and G2-P in the indicated solvents. λex = 375 nm, $c = 10^{-3}$ M.

Table 1
Fluorescence lifetime and quantum yield data for phosphonate-functionalized siloxane dendritic macromolecules G0-P-G2-P.

Compound	τ_1/ns	τ_1%	τ_1/ns	τ_1%	τ/ns	x^2	QY/% (λ_{ex} = 375 nm)
G0-P	1.3415	86.51	5.5160	13.49	1.90	1.174	4.67
G1-P	2.0724	57.85	5.3375	42.15	3.45	1.308	12.32
G2-P	2.0267	21.51	7.5975	78.49	6.40	0.996	22.14

Note: Tests were conducted under liquid conditions.

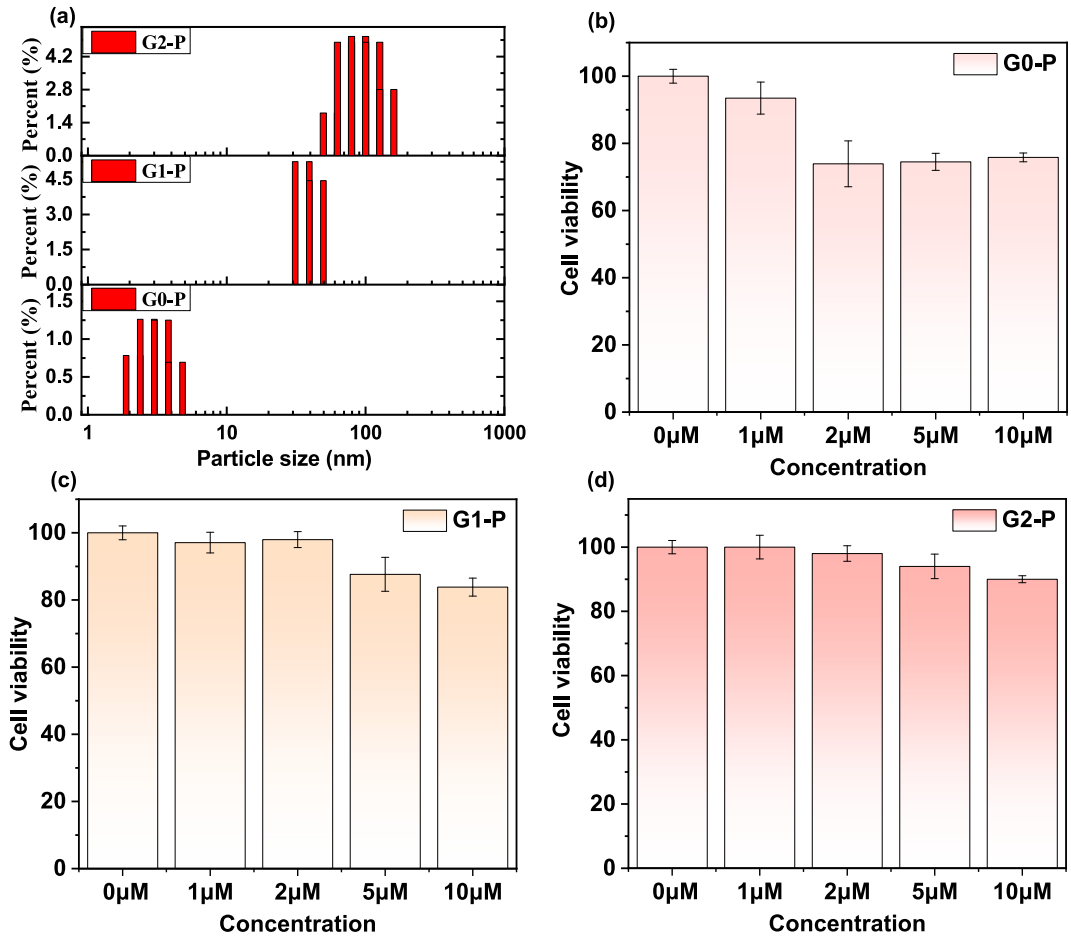


Fig. 5. (a) DLS data of phosphonate-functionalized siloxane dendrimers G0-P-G2-P; Using cck-8 to test the effect of different concentrations of materials on HeLa cell-activated animals. (b) G0-P, (c) G1-P, (d) G2-P. (*p < 0.05, **p < 0.01, ***p < 0.001, n = 3).

This morphological transition (dispersed → nanoparticles → networks) confirms that poor solvent conditions promote larger aggregate formation in higher-generation dendrimers, which directly correlates with both increased hydrodynamic dimensions and improved fluorescence quantum yields – key characteristics of AIE behavior.

To elucidate the structure–property relationship of the fluorophores, we conducted systematic investigations into their intrinsic luminescence mechanisms through quantum chemical calculations. The frontier molecular orbitals, including the highest occupied molecular orbital (HOMO) and lowest unoccupied molecular orbital (LUMO), were computationally analyzed to understand their electronic properties. Figs. S25-S28 present the optimized molecular geometries and corresponding HOMO-LUMO distributions obtained using Gaussian 16 software with the long-range corrected CAM-B3LYP functional and 6-311G (d) basis set. Frequency analyses confirmed all optimized structures corresponded to true minima on the potential energy surfaces.

The calculated HOMO-LUMO energy gaps (ΔE) revealed significant differences between compounds: 7.86 eV for G0 and 7.23 eV for G1. Remarkably, phosphonate-functionalized dendrimers (G0-P and G1-P) exhibited substantially reduced energy gaps of 6.34 eV and 6.04 eV, respectively. This 19.3 % and 16.5 % reduction in ΔE demonstrates that phosphonate functionalization effectively modulates the electronic structure, leading to optimized photophysical properties through stabilization of both HOMO and LUMO energy levels. Detailed charge distribution analysis (Figs. S29-S30, Table S1) revealed that while phosphorus atoms maintain comparable charge densities, the ethoxylate oxygen atoms mediate more homogeneous charge delocalization throughout the molecular framework, facilitating efficient intramolecular charge transfer and contributing to the observed photophysical enhancements. Structural analysis of P-C bond distances (Figs. S31-S32, Table S2) showed significantly shortened bond lengths (1.84 Å for G0-P, 1.89 Å for G1-P) compared to the sum of van der Waals radii (3.50 Å), indicating strong orbital overlap and efficient TSC that enables effective exciton delocalization within the phosphonate architecture, ultimately enhancing fluorescence emission.

Finally, cytotoxicity was evaluated across varying concentrations. As

shown in Fig. 5b-5d, increasing the material concentration did not significantly impact cell viability. Remarkably, G2-P exhibited excellent cytocompatibility, and had little effect on cells even at higher concentrations. These results suggest that G2-P supports cell growth in a culture environment with essentially no adverse effects. In summary, the synthesized materials display exceptional biocompatibility, establishing a strong foundation for further research in biological and biomedical applications.

3. Conclusion

In summary, a novel series of phosphonate-functionalized siloxane dendritic macromolecules (G0-P to G2-P) exhibiting AIE behavior was successfully synthesized using a controlled dispersion method with a phosphate ester core. The AIE properties were comprehensively characterized using UV–visible and fluorescence spectroscopy, quantum yield measurements, DLS, and TEM, revealing distinct generation-dependent AIE effects. The AIE phenomenon intensified with increasing dendrimer generation. Notably, DLS size growth (G0-P → G2-P) together with fluorescence lifetime extension (Table 1) confirms that generational growth enhances the TSC effect through spatial constraints rather than mere aggregation leading to enhanced luminescence. Furthermore, cytotoxicity assays demonstrated excellent biocompatibility across all generations. This study not only broadens the range of phosphorus-containing dendrimeric macromolecules but also introduces a promising strategy for designing luminescent biomass-based functional materials.

CRediT authorship contribution statement

Zhihui Yang: Writing – original draft, Visualization, Methodology, Investigation. **Fanghao Sun:** Visualization, Validation, Investigation, Formal analysis. **Weizhi Tian:** Methodology, Investigation. **Haobo Shi:** Investigation. **Haifeng Lu:** Writing – review & editing, Resources, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

We thank the National Key·R&D·Program of China·(2022YFF1400300) supported our work.

Appendix A. Supplementary data

Figure S1-S20: Product Nuclear Magnetic Resonance (NMR) spectra (marked in order in the supplementary document). Figure S21: UV visible absorption spectra of three target compounds. Transmission electron microscopy (TEM) images of G0, G1, and G2 in Figures S22-S24. Figures S25-S32 and Tables S1-S2: Theoretical calculation analysis and related datasets. Supplementary data to this article can be found online at https://doi.org/10.1016/j.eurpolymj.2025.113952.

Data availability

Data will be made available on request.

References

[1] J.L. Hu, B.J. Zhao, R.R. Wen, X.B. Zhang, Y.X. Zhang, D.S. Kohane, Q. Liu, Squaraine dye-sensitized upconversion with enhanced stability and minimized aggregation-caused quenching, Nano Lett. 23 (2023) 5209–5216.
[2] Y.Y. Liu, X. Zhang, K. Li, Q.C. Peng, Y.J. Qin, H.W. Hou, S.Q. Zang, B.Z. Tang, Restriction of intramolecular vibration in aggregation-induced emission luminogens: applications in multifunctional luminescent metal–organic frameworks, Angew. Chem. Int. Ed. 60 (2021) 22417.
[3] H.S. He, C. Liu, J. Ming, Y.J. Lv, J.P. Qi, Y. Lu, X.C. Dong, W.L. Zhao, W. Wu, Accurate and sensitive probing of onset of micellization based on absolute aggregation-caused quenching effect, Aggregate 3 (2022) e163.
[4] P. Chowdhury, A. Banerjee, B. Saha, K. Bauri, P. De, Stimuli-responsive aggregation-induced emission (AIE)-active polymers for biomedical applications, ACS Biomater. Sci. Eng. 8 (2022) 4207–4229.
[5] K.T. Prakash, N. Singh, V. Venkatesh, Synthesis of novel luminescent copper nanoclusters with substituent driven self-assembly and aggregation induced emission (AIE), Chem Commun. 55 (2019) 322–325.
[6] Sun, H. Li, X. Gu, B.Z. Tang, Photoactivatable biomedical materials based on luminogens with aggregation-induced emission (AIE) characteristics, Adv. Healthc. Mater. 10 (2021) 2101177.
[7] M.M. Islam, Z. Hu, Q.S. Wang, C. Redshaw, X. Feng, Pyrene-based aggregation-induced emission luminogens and their applications, Mater. Chem. Front. 3 (2019) 762–781.
[8] J. Yang, M. Fang, Z. Li, Organic luminescent materials: the concentration on aggregates from aggregation-induced emission, Aggregate 1 (2020) 6–18.
[9] F. Zhang, H. Xie, B. Guo, C.Z. Zhu, J. Xu, AIE-active macromolecules: designs, performances, and applications, Polym. Chem. 13 (2022) 8–43.
[10] V.K. Gawade, R.W. Jadhav, S.V. Bhosale, AIE-based & organic luminescent materials: nanoarchitectonics and advanced applications, Chem. Asian J. 19 (2024) e202400682.
[11] G. Ahumada, M. Borkowska, Fluorescent polymers conspectus, Polymers 14 (2022) 1118.
[12] M. Asad, M.I. Anwar, A. Abbas, A. Younas, S. Hussain, R.X. Gao, L.K. Li, M. Shahid, S. Khan, AIE based luminescent porous materials as cutting-edge tool for environmental monitoring: state of the art advances and perspectives, Coord. Chem. Rev. 463 (2022) 214539.
[13] H.X. Liu, H.T. Bai, J.W.Y. Lam, R.T.K. Kwok, B.Z. Tang, Recent advances in aggregation-induced emission materials for enhancing solar energy utilization, Nanoscale Horiz. 8 (2023) 453–459.
[14] Y.R. Liu, Y.L. Li, H. Liu, S.S. Yu, S.H. Ma, L.B. Xing, F. Zhou, High-strength anisotropic fluorescent hydrogel based on solvent exchange for patterning, ACS Appl. Mater. Inter. 17 (2025) 4114–4123.
[15] Z. Wang, J. Ma, C. Li, H. Zhang, Conjugated aggregation-induced fluorescent materials for biofluorescent probes: a review, Biosensors. 13 (2023) 159.
[16] S. Gouthaman, A. Jayaraj, M. Sugunalakshmi, G. Sivaraman, P. Chinna Ayya Swamy, Supramolecular self-assembly mediated aggregation-induced emission of fluorene-derived cyanostilbenes: multifunctional probes for live cell-imaging, J. Mater. Chem. B 10 (2022) 2238–2250.
[17] M.E. Gomes, R.M.A. Domingues, Magnetic systems for regenerative medicine, Adv. Funct. Mater. 32 (2022) 2211925.
[18] D. Wang, J. Chen, Y.L. Li, S.H. Feng, Solid state chemistry and inorganic synthetic chemistry-novel structures and accurate syntheses of inorganic materials, Small Methods. 9 (2024).
[19] S.Q. Chen, B.W. Li, B. Liu, AIEgen-biomacromolecule conjugates: visualized delivery and light-controlled theranostic platforms, J. Control. Release 378 (2025) 605–618.
[20] Y.B. Feng, H.X. Yan, F. Ding, T. Bai, Y.F. Nie, Y. Zhao, W.X. Feng, B.Z. Tang, Multiring-induced multicolour emission: hyperbranched polysiloxane with silicon bridge for data encryption, Mater. Chem. Front. 4 (2020) 1375–1382.
[21] L.F. Xu, X. Liang, S.L. Zhong, Y. Gao, X.J. Cui, Seeking brightness from nature: Sustainable AIE macromolecule with clustering-triggered emission of xanthan gum and its multiple applications, Colloids Surf. B 206 (2021) 111961.
[22] F.X. Chen, Y.X. Jin, J. Luo, L.Z. Wei, B.L. Jiang, S. Guo, C. Wei, Y.Y. Gong, Poly-L-aspartic acid based nonconventional luminescent biomacromolecules with efficient emission in dilute solutions for Al^{3+} detection, Int. J. Biol. Macromol. 226 (2023) 1387–1395.
[23] A. Dutta, M. Mahapatra, M. Mitra, A. Banerjee, N.N. Ghosh, P.K. Chattopadhyay, D. K. Maiti, N.R. Singha, Nonconventional biocompatible macromolecular AEEgens for sensitive detections and removals of Cu (II) and Fe (III): N and/ or O donor (s) selective coordinations of metal ions, Sensor. Actuat. B-Chem. 331 (2021) 129386.
[24] H.Y. Wang, B. Aydıner, Z. Seferoglu, F. Bureš, J.L. Liu, Nonconventional biocompatible macromolecular AEEgens for sensitive detections and removals of Cu(II) and Fe(III): N and/ or O donor(s) selective coordinations of metal ions, Dyes Pigm. 205 (2022) 110354.
[25] N. Jiang, C.Y. Zhu, K.X. Li, Y.H. Xu, M.R. Bryce, Recent progress in nonconventional luminescent macromolecules and their applications, Macromolecules 57 (2024) 5561–5577.
[26] R. Hu, J. Wang, A.J. Qin, B.Z. Tang, Aggregation-induced emission-active biomacromolecules: progress, challenges, and opportunities, Biomacromolecules 23 (2022) 2185–2196.
[27] Y. Wang, Z. Zhao, W.Z. Yuan, Intrinsic luminescence from nonaromatic biomolecules, Chem. plus Chem. 85 (2020) 1065–1080.
[28] K. Dueñas-Parro, O. Gulias, M. Agut, F. de la de la Cruz-Martínez, A. Lara-Sánchez, J.A. Castro-Osma, J.F. García-Reyes, A. Sánchez-Ruiz, C. Martín, S. Nonell, R. Bresolí-Obach, Cluster and kill: the use of clustering-triggered emission materials for singlet oxygen photosensitization in antimicrobial photodynamic therapy, Adv. Optical Mater. 13 (2024) 2402179.
[29] Y. Li, Y.L. Zhu, X.C. Li, P.A. Chen, Unexpected full-color luminescence produced from the aggregation of unconventional chromophores in novel polyborosilazane dendrimers, Chem. Sci. 15 (2024) 17988–17999.

[30] T.J. Yang, Y.X. Li, Z.H. Zhao, W.Z. Yuan, Clustering-triggered phosphorescence of nonconventional luminophores, Sci. China Chem. 66 (2023) 367–387.
[31] D.A. Tomalia, B.K. Maculewicz, K.-A.-M. Johnson, H.F. Brinkman, A. Janaszewska, D.M. Hedstrand, Prog. Polym. Sci. 90 (2019) 35–117.
[32] M. Studzian, Ł. Pułaski, D.A. Tomalia, B.K. Maculewicz, Non-traditional intrinsic luminescence (NTIL): dynamic quenching demonstrates the presence of two distinct fluorophore types associated with NTIL Behavior in Pyrrolidone-Terminated PAMAM Dendrimers, J. Phys. Chem. C 123 (2019) 18007–18016.
[33] O. Varnavski, R.G. Ispasoiu, L. Balogh, D. Tomalia, T. Goodson, Ultrafast time-resolved photoluminescence from novel metal–dendrimer nanocomposites, J. Chem. Phys. 114 (2001) 1962–1965.
[34] P.J. Pacheco-Linan, A. Garzon, J. Tolosa, I. Bravo, J. Canales-Vazquez, J. Rodriguez-Lopez, J. Albaladejo, J.C. Garcia-Martinez, pH-sensitive fluorescence lifetime molecular probes based on functionalized tristyrylbenzene, J. Phys. Chem. c. 120 (2016) 18771–18779.
[35] J.H. Gao, B.Q. Wan, M.S. Zheng, J.W. Zha, Citric acid-induced room temperature self-healing polysiloxane elastomers with tunable mechanical properties and untraditional AIE fluorescence, Polym. Chem. 13 (2022) 5412–5421.
[36] H.K. Zhang, Z. Zhao, P.R. McGonigal, R.Q. Ye, S.J. Liu, J.W.Y. Lam, R.T.K. Kwok, W.Z. Yuan, J.P. Xie, A.L. Rogach, B.Z. Tang, Clusterization-triggered emission: Uncommon luminescence from common materials, Mater. Today 32 (2020) 275.
[37] T. Bai, H.X. Yan, Y.S. Zhang, L.L. Guo, Y. Zhao, Disulfide-containing polysiloxane with multicolor fluorescence and visible-light excitation: design, synthesis, and mechanism exploration, J. Phys. Chem. B 125 (2021) 4321–4329.
[38] D.A. Tomalia, B. Klajnert-Maculewicz, K.A.M. Johnson, H.F. Brinkman, A. Janaszewska, D.M. Hedstrand, Non-traditional intrinsic luminescence: inexplicable blue fluorescence observed for dendrimers, macromolecules and small molecular structures lacking traditional/conventional luminophores, Prog. Polym. Sci. 90 (2019) 35–117.
[39] Z. Zhao, H. Zhang, J.W.Y. Lam, B.Z. Tang, Aggregation-induced emission: new vistas at the aggregate level, Angew. Chem., Int. Ed. 59 (2020) 9888–9907.
[40] W. Zhao, Z. He, B.Z. Tang, Room-temperature phosphorescence from organic aggregates, Nat. Rev. Mater. 5 (2020) 869–885.
[41] Y. Zhao, J. Long, P. Zhuang, Y. Ji, C. He, H. Wang, Transforming polyethylene and polypropylene into nontraditional fluorescent polymers by thermal oxidation, J. Mater. Chem. C 10 (2022) 1010–1016.
[42] Z. Zhou, X. Chen, Y. Wang, C. Hu, T. Li, S. Wang, W. Dong, J. Qiao, Branched copolymers with tunable clusteroluminescence in high quantum yield, ACS Macro Lett. 12 (2023) 1523–1529.
[43] F. Kausar, Z. Zhao, T. Yang, W. Hou, Y. Li, Y. Zhang, W.Z. Yuan, Michael polyaddition approach towards sulfur enriched nonaromatic polymers with fluorescence-phosphorescence dual emission, Macromol. Rapid Commun. 42 (2021) 2100036.
[44] Y. Joo, V. Agarkar, S.H. Sung, B.M. Savoie, B.W. Boudouris, A nonconjugated radical polymer glass with high electrical conductivity, Science 359 (2018) 1391–1395.
[45] D.A. Tomalia, B.K. Maculewicz, K.-A.-M. Johnson, H.F. Brinkman, A. Janaszewska, D.M. Hedstrand, Non-traditional intrinsic luminescence: inexplicable blue fluorescence observed for dendrimers, macromolecules and small molecular structures lacking traditional/conventional luminophores, Prog. Polym. Sci. 90 (2019) 35–117.
[46] X.N. Li, M. Li, M.Q. Yang, H.N. Xiao, L.D. Wang, Z.J. Chen, S.X. Liu, J. Li, S.J. Li, T. D. James, "Irregular" aggregation-induced emission luminogens, Coord. Chem. Rev. 418 (2020) 213358.
[47] Y.Q. Du, Y.W. Liu, J.W. Li, Y.Y. He, Y.B. Li, H.X. Yan, Nonconventional luminescent piperazine-containing hyperbranched polysiloxanes with pure n-electron, Small 19 (2023) 2302095.
[48] L.T. Canham, A. Loni, P.D.J. Calcott, A.J. Simons, C. Reeves, M.R. Houlton, J. P. Newey, K.J. Nash, T.I. Cox, On the origin of blue luminescence arising from atmospheric impregnation of oxidized porous silicon, Thin Solid Films 276 (1996) 112.
[49] L.D. Carlos, R.A. Sá Ferreira, R.N. Pereira, M. Assunção, V.D.Z. Bermudez, White-light emission of amine-functionalized organic/inorganic hybrids: emitting centers and recombination mechanisms, J. Phys. Chem. B 108 (2004) 14924.
[50] Y. Zhao, L. Xu, Y.Y. He, Z.X. Feng, W.X. Feng, H.X. Yan, Nonconventional aggregation-induced emission polysiloxanes: Structures, characteristics, and applications, Aggregate 5 (2024) e471.
[51] H. K. Zhang, Z. Zhao, P. R. McGonigal, R. Q. Ye, S. J. Liu, J. W. Y. Lam, R. T. K. Kwok, W. Z. Yuan, J. P. Xie, A. L. Rogach, B. Z. Tang, Clusterization-triggered emission: Uncommon luminescence from common materials, Mater. Today 32 (2020) 32, 275.
[52] Z.H. Zhao, A. Li, W.Z. Yuan, Nonconventional luminophores: emission mechanism, regulation, and applications, Acc. Chem. Res. 58 (2025) 612–624.
[53] X. Chen, Y. Sun, W.Z. Yuan, Nonconventional luminescent polymers: emission regulation and applications, Macromol. Chem. Phys. 226 (2025) 2400401.
[54] H. Lu, L.L. Feng, S.S. Li, J. Zhang, H.F. Lu, S.Y. Feng, Unexpected strong blue photoluminescence produced from the aggregation of unconventional chromophores in novel siloxane-poly (amidoamine) dendrimers, Macromolecules 48 (2015) 476–482.
[55] X.Q. Wang, W. Wang, W.J. Li, Y. Qin, G.Q. Yin, W.L. Jiang, X.P. Li, S. Wu, H. B. Yang, Rotaxane-branched dendrimers with aggregation-induced emission behavior, Org. Chem. Front. 6 (2019) 1686–1691.